



Top Nine Open Source Cloud Management Platforms

A cloud management platform (CMP) is a comprehensive software suite with integrated tools that an enterprise can use to monitor and control cloud computing resources. There are many CMPs out there, both open source and proprietary. This article explores how cloud platforms work, their capabilities, as well as a selection of the top open source cloud platforms of the day



Anand Nayyar
works at Duy Tan University in Vietnam. He loves to explore open source technologies, sensor communications, network security and the Internet of Things

In recent times, all the enterprises that have started or have adopted cloud computing face new challenges with regard to ensuring the same visibility in cloud applications as they have with their on-premise apps. With any cloud-based full implementation, the issues that users face include preserving the integrity, usability and security of all the data migrated online. According to John Webster, senior partner and analyst for the Evaluator Group, and whose research area is the hybrid cloud and Big Data, “As enterprise IT operations start to expand our capabilities and resources into the cloud, we now want to manage cloud resources with the same policy, procedure, guidance and expectations that we have over our existing IT environment.”

The only solution to all the issues arising from cloud-based implementations is a cloud management platform (CMP). This provides a rich set of capabilities for discovery, template-based provisioning, orchestration and automation. It also enables operational monitoring and management, governance and cost optimisation across multiple public and private clouds as well as virtual and bare-metal servers.

A CMP is a comprehensive suite with integrated software tools that an enterprise can use to monitor and control cloud computing resources. An enterprise can use a CMP for either a private or public cloud, but CMPs facilitate toolsets for hybrid and multi-cloud models to centralise the control of various cloud-based infrastructures.

A wide range of tools is avail-

able to deal with these challenges and help companies efficiently operate applications and services in the cloud. Vendors offer a variety of cloud management tools that enable IT organisations to build, purchase, manage, monitor, track and optimise their cloud resources. With the help of these tools, organisations can save time and effort while allowing IT staff to focus on more strategic goals. These tools also help in monitoring users’ interactions with the cloud infrastructure and in managing resource allocation.

In the cloud marketplace, different tools have unique feature sets. Capabilities include unified management across multiple clouds, integration with third-party tools for configuration and monitoring purposes, dashboards and reports for detailed information about resource consumption, notifications and alerts when predefined thresholds are reached, and controlled access to resources to avoid over usage or unauthorised access. **By simplifying the management of cloud environments and reducing the complexity and cost of managing multiple activities, the right cloud management tool can bring speed, flexibility, security and cost efficiency to any organisation.**

So the important question is, “Which CMP is the best?” Well, there are both commercial and open source options. Nowadays, open source technology has become a central facet of cloud computing for many users around the world. In this article, we discuss the top open source cloud management platforms to assist admins, systems analysts, network security as well as cloud computing professionals select the best platform, based on the requirements of the enterprise.

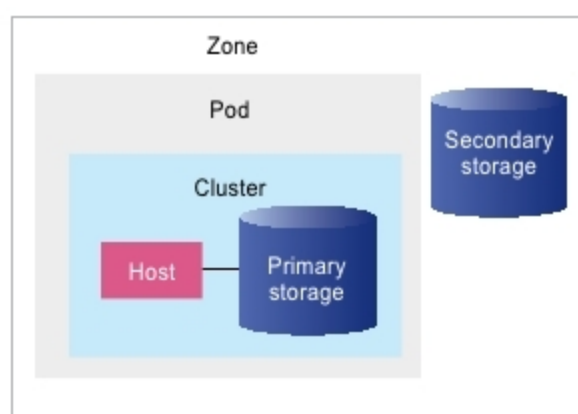


Fig. 1: CloudStack architecture